

CHAPTER 5.2A: REQUIREMENTS FOR PRIVACY MECHANISMS

FOUNDATIONS OF DISTRIBUTED SYSTEMS

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We want to answer a few important questions:

- What has gone wrong with privacy mechanisms in the past?
- What can we learn from that when designing modern privacy mechanisms?

1. Problem #1: Naïve anonymization

2. Problem #2: Privacy by obscurity

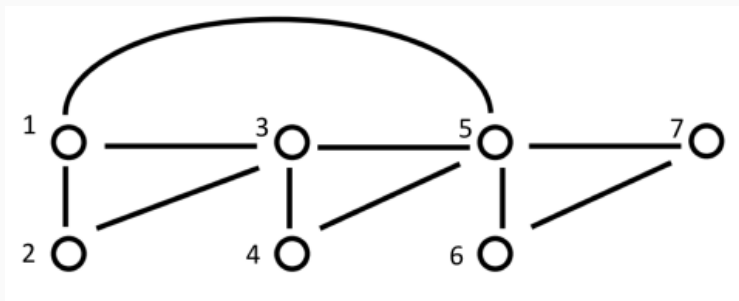
3. Problem #3: Multiple releases

4. Problem #4: Post-processing

1. PROBLEM #1: NAÏVE ANONYMIZATION

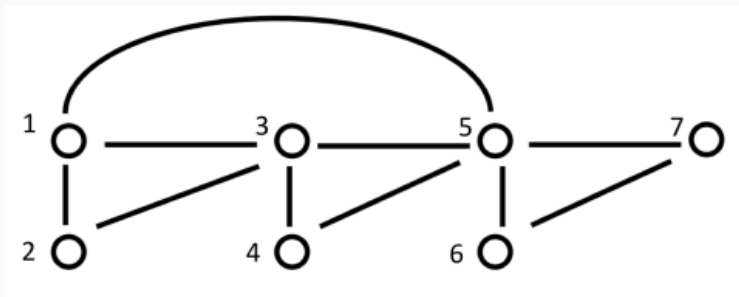
SOCIAL GRAPH ANONYMIZATION

- Social graph: each node represents an individual, each edge represents some kind of relationship
- For example email communications graph: edges represent email sending
- Is it safe to release this graph?

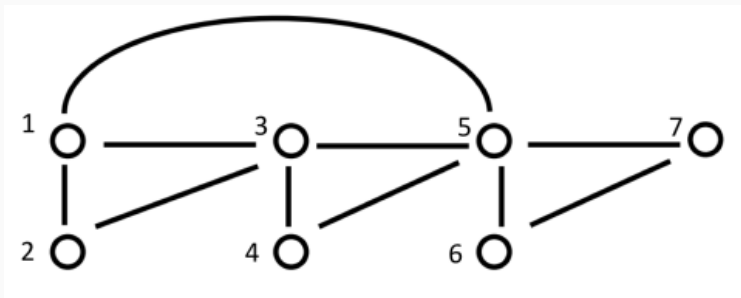


SOCIAL GRAPH ANONYMIZATION

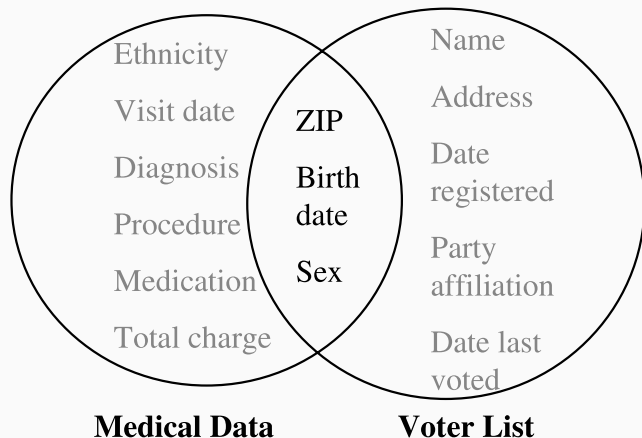
- What if you know that:
 - You (Alice) and your friend Cathy are in the dataset
 - You have sent emails to Bob, Cathy, and Ed
 - Cathy has sent emails to everyone except Ed



SOCIAL GRAPH ANONYMIZATION



- Node 1 is Alice (only one node with degree=3)
- Node 5 is Cathy (only one node with degree=5)
- Node 3 is Bob (only Bob sent emails to both Alice and Cathy)
- Node 2 is Ed (Alice only sent emails to Bob, Cathy, and Ed)



¹L. Sweeney, "K-Anonymity: A Model for Protecting Privacy," *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 10, no. 05, pp. 557–570, Oct. 2002. DOI: 10.1142/S0218488502001648. [Online]. Available: <http://www.worldscientific.com/doi/abs/10.1142/S0218488502001648>.

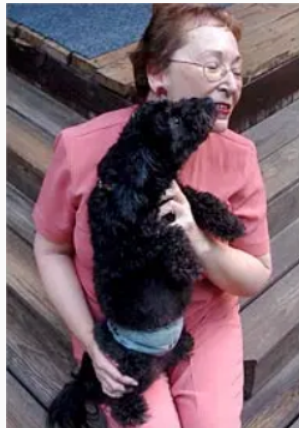
AOL SEARCH HISTORY

- In 2006, AOL published 20 million search queries from 650,000 users
- User identifiers were replaced with random numbers, so data was considered anonymized
- 5 days later:

The New York Times

A Face Is Exposed for AOL Searcher No. 4417749

Source: <https://www.nytimes.com/2006/08/09/technology/09aol.html>



AOL SEARCH HISTORY²

2178	pergola	2006-04-08 18:58:22	28		http://www.plansnow.com
2178	google	2006-04-11 11:06:41			
2178	irs publication 505	2006-04-11 22:19:49	2		http://www.irs.gov
2178	prestonandsteve	2006-04-12 09:11:37	1		http://www.prestonandsteve.com
2178	msn.com	2006-04-12 16:02:05	1		http://www.msn.com
2178	mnf-iraq.mil	2006-04-13 20:59:59			
2178	army.mil	2006-04-13 21:03:22			
2178	us constitution center	2006-04-19 22:06:10	1		http://www.constitutioncenter.org
2178	philadelphia rocky statue	2006-04-19 22:12:42	7		http://uk.holidaysguide.yahoo.com
2178	philadelphia art museum	2006-04-19 22:15:23	1		http://www.philamuseum.org
2178	us constitution center	2006-04-19 22:17:44			
2178	philadelphia independence mall	2006-04-19 22:18:19	6		http://www.independencevisitorcenter.com
2178	babiesrus	2006-04-20 19:35:25	1		http://www.babiesrus.com
2178	securespeed.us	2006-04-22 09:10:00			
2178	lily pads breast	2006-04-29 14:52:29			
2178	army.mil	2006-05-03 09:17:05			
2178	boppy covers	2006-05-08 13:14:00			
2178	ebay.com	2006-05-08 13:22:50			
2178	once upon a child	2006-05-09 09:55:46	2		http://eelement.inetium.com
2178	scott honda	2006-05-09 13:56:50			
2178	braxton hicks contractions	2006-05-09 17:08:21	7		http://www.pregnancy-info.net
2178	foods to avoid when pregnant	2006-05-09 19:32:42	4		http://www.americanpregnancy.org
2178	foods to avoid when breast feeding	2006-05-09 19:36:44	4		http://www.babycenter.com
2178	infant innocation safety	2006-05-09 19:41:32			
2178	hepatitis b vaccine safety infants	2006-05-09 19:43:37	2		http://www.vaccinesafety.edu
2178	roth ira calculator	2006-05-09 19:53:35	5		http://www.statefarm.com

²<https://archive.org/details/aolsearchdata2006>

AOL SEARCH HISTORY



— IN SOLIDARITY WITH THE MANY AOL USERS WHOSE OFTEN EMBARRASSING WEB SEARCHES WERE RELEASED TO THE PUBLIC, I OFFER A SAMPLE OF MY OWN SEARCH HISTORY:



Web [Images](#) [Video](#) ^{New!} [News](#) [Maps](#) [more »](#)

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[Language Tools](#)

velociraptors
site:imdb.com 'jurassic park'
raptors
dromaeos aurids
utahraptor
'home depot' deadbolts
security home improvement
surviving a raptor attack
robert bakker paleontologist
robert bakker 'possible raptor sympathizer'
site:en.wikipedia.org surviving a raptor attack
learning from mistakes in jurassic park
big-game rifles
tire irons
treating raptor wounds
do raptors fear fire
how to make a molotov cocktail
do raptors fear death
can raptors pick locks
how to tell if my neighbors are raptors

Source: <https://xkcd.com/155/>

SIMPLY REMOVING IDENTIFIERS IS NOT ENOUGH!

- Adversary has additional knowledge if they are in the dataset
- Adversary may be able to purchase overlapping data
- Data items may reveal identifying information

Requirement #1: Resilience to background knowledge

A privacy mechanism must be able to protect individuals' privacy from attackers who may possess background knowledge

2. PROBLEM #2: PRIVACY BY OBSCURITY

- The email provider from above wants to release customer ages in addition to the social graph
- They perturb the ages using a linear function $ax + b$, with secret parameters a and b
- Is it safe to release this table?

node id	age
1	40
2	34
3	52
4	28
5	48
6	22
7	92

- What if you know that Alice's real age is 25 and Cathy's is 29?

node id	age
1	40
2	34
3	52
4	28
5	48
6	22
7	92

DATA PERTURBATION WITH SECRET PARAMETERS

- What if you know that Alice's real age is 25 and Cathy's is 29?

$$25a + b = 40$$

$$29a + b = 49$$

$$a = 2$$

$$b = -10$$

node id	name	age	real age
1	Alice	40	25
2	Ed	34	
3	Bob	52	
4		28	
5	Cathy	48	29
6		22	
7		92	

- What if you know that Alice's real age is 25 and Cathy's is 29?

$$25a + b = 40$$

$$29a + b = 49$$

$$a = 2$$

$$b = -10$$

node id	name	age	real age
1	Alice	40	25
2	Ed	34	22
3	Bob	52	31
4		28	19
5	Cathy	48	29
6		22	16
7		92	51

- In 1883, Dutch cryptographer August Kerckhoff published design principles for military cryptography
- Principle 2: “It should not require secrecy, and it should not be a problem if it falls into enemy hands”
- Modern cryptography follows this principle: mechanisms must be secure, even if:
 - The algorithm/protocol is public
 - The parameters are public
 - Only random bits used during computations (“keys”) can be assumed secret
- Benefit: scientific study of algorithms only possible if they are open, so openness leads to better algorithms

KEEPING ALGORITHMS AND PARAMETERS SECRET IS NOT ENOUGH!

- Adversary is smart: just because we cannot infer sensitive information from the output of a privacy mechanism, doesn't mean that the adversary cannot
- Other researchers are smart: they can critique privacy mechanisms if details are public, thereby help to improve the mechanism

Requirement #2: Privacy without obscurity

Attacker must be assumed to know the algorithm used as well as all parameters

3. PROBLEM #3: MULTIPLE RELEASES

INDEPENDENT DATA RELEASES³

- Two hospitals want to release anonymized patient records
- They have removed identifiers, generalized some attributes (age), omitted others (nationality)
- Is it safe to release this data?

	Non-Sensitive			Sensitive
	Zip code	Age	Nationality	Condition
1	130**	<30	*	AIDS
2	130**	<30	*	Heart Disease
3	130**	<30	*	Viral Infection
4	130**	<30	*	Viral Infection
5	130**	≥40	*	Cancer
6	130**	≥40	*	Heart Disease
7	130**	≥40	*	Viral Infection
8	130**	≥40	*	Viral Infection
9	130**	3*	*	Cancer
10	130**	3*	*	Cancer
11	130**	3*	*	Cancer
12	130**	3*	*	Cancer

	Non-Sensitive			Sensitive
	Zip code	Age	Nationality	Condition
1	130**	<35	*	AIDS
2	130**	<35	*	Tuberculosis
3	130**	<35	*	Flu
4	130**	<35	*	Tuberculosis
5	130**	<35	*	Cancer
6	130**	<35	*	Cancer
7	130**	≥35	*	Cancer
8	130**	≥35	*	Cancer
9	130**	≥35	*	Cancer
10	130**	≥35	*	Tuberculosis
11	130**	≥35	*	Viral Infection
12	130**	≥35	*	Viral Infection

³S. R. Ganta, S. P. Kasiviswanathan, and A. Smith, "Composition attacks and auxiliary information in data privacy," in *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD '08, New York, NY, USA: Association for Computing Machinery, Aug. 2008, pp. 265–273. DOI: 10.1145/1401890.1401926. [Online]. Available: <https://doi.org/10.1145/1401890.1401926>.

- What if you are Alice's employer and know that she is 28 and has visited both hospitals?

	Non-Sensitive			Sensitive
	Zip code	Age	Nationality	Condition
1	130**	<30	*	AIDS
2	130**	<30	*	Heart Disease
3	130**	<30	*	Viral Infection
4	130**	<30	*	Viral Infection
5	130**	≥40	*	Cancer
6	130**	≥40	*	Heart Disease
7	130**	≥40	*	Viral Infection
8	130**	≥40	*	Viral Infection
9	130**	3*	*	Cancer
10	130**	3*	*	Cancer
11	130**	3*	*	Cancer
12	130**	3*	*	Cancer

	Non-Sensitive			Sensitive
	Zip code	Age	Nationality	Condition
1	130**	<35	*	AIDS
2	130**	<35	*	Tuberculosis
3	130**	<35	*	Flu
4	130**	<35	*	Tuberculosis
5	130**	<35	*	Cancer
6	130**	<35	*	Cancer
7	130**	≥35	*	Cancer
8	130**	≥35	*	Cancer
9	130**	≥35	*	Cancer
10	130**	≥35	*	Tuberculosis
11	130**	≥35	*	Viral Infection
12	130**	≥35	*	Viral Infection

SUCCESSIVE DATA RELEASES

- A hospital wants to release data quarterly, so that each release includes data from the past 6 months
- They take care to generalize age and zip code, so that no individual is uniquely identifiable in each release
- Is it safe to release this data?

Age	Zip.	Disease	Age	Zip.	Disease
[21, 22]	[12k, 14k]	dyspepsia	[21, 23]	[12k, 25k]	dyspepsia
[21, 22]	[12k, 14k]	bronchitis	[21, 23]	[12k, 25k]	gastritis
[23, 24]	[18k, 25k]	flu	[25, 43]	[21k, 33k]	flu
[23, 24]	[18k, 25k]	gastritis	[25, 43]	[21k, 33k]	dyspepsia
[36, 41]	[20k, 27k]	flu	[25, 43]	[21k, 33k]	gastritis
[36, 41]	[20k, 27k]	gastritis	[41, 46]	[20k, 30k]	flu
[37, 43]	[26k, 35k]	dyspepsia	[41, 46]	[20k, 30k]	gastritis
[37, 43]	[26k, 35k]	flu	[54, 56]	[31k, 34k]	dyspepsia
[37, 43]	[26k, 35k]	gastritis	[54, 56]	[31k, 34k]	gastritis
[52, 56]	[33k, 34k]	dyspepsia	[60, 65]	[36k, 44k]	gastritis
[52, 56]	[33k, 34k]	gastritis	[60, 65]	[36k, 44k]	flu

SUCCESSIVE DATA RELEASES⁴

- What if you know that Bob, who is 21 and lives in zip code 12000, has been to the hospital around 4 or 5 months ago? (i.e., Bob is in both data releases)

Age	Zip.	Disease	Age	Zip.	Disease
[21, 22]	[12k, 14k]	dyspepsia	[21, 23]	[12k, 25k]	dyspepsia
[21, 22]	[12k, 14k]	bronchitis	[21, 23]	[12k, 25k]	gastritis
[23, 24]	[18k, 25k]	flu	[25, 43]	[21k, 33k]	flu
[23, 24]	[18k, 25k]	gastritis	[25, 43]	[21k, 33k]	dyspepsia
[36, 41]	[20k, 27k]	flu	[25, 43]	[21k, 33k]	gastritis
[36, 41]	[20k, 27k]	gastritis	[41, 46]	[20k, 30k]	flu
[37, 43]	[26k, 35k]	dyspepsia	[41, 46]	[20k, 30k]	gastritis
[37, 43]	[26k, 35k]	flu	[54, 56]	[31k, 34k]	dyspepsia
[37, 43]	[26k, 35k]	gastritis	[54, 56]	[31k, 34k]	gastritis
[52, 56]	[33k, 34k]	dyspepsia	[60, 65]	[36k, 44k]	gastritis
[52, 56]	[33k, 34k]	gastritis	[60, 65]	[36k, 44k]	flu

⁴X. Xiao and Y. Tao, "M-invariance: Towards Privacy Preserving Re-publication of Dynamic Datasets," in *ACM SIGMOD International Conference on Management of Data*, ser. SIGMOD '07, Beijing, China: ACM, Jun. 2007, pp. 689–700. DOI: 10.1145/1247480.1247556. [Online]. Available: <http://doi.acm.org/10.1145/1247480.1247556>.

INDEPENDENT APPLICATIONS OF PRIVACY MECHANISMS ARE NOT ENOUGH!

- Adversary may have data that was independently anonymized & released by others
- We may want to periodically update our published data (adversary will hold on to any data that was published at some point)

Requirement #3: Composition over multiple releases

Allow a graceful degradation of privacy with multiple invocations on the same data

4. PROBLEM #4: POST-PROCESSING

COMPUTING ON ANONYMIZED DATA⁶

- Suppose a healthcare organization has released the following table about how many people with a certain disease were discharged from hospital in a particular year, depending on their age and ethnicity⁵
- Values smaller than 11 were suppressed to protect privacy
- Is it safe to release this data?

		total # discharges	Race/ ethnicity					
			White	Black	Hispanic	Asian/ Pacific Islander	Native American	Other
total# discharges		735	535	82	58	18	*	19
Age group	1-17	*	*	*	*	*	*	*
	18-44	70	40	13	*	*	*	*
	45-64	330	236	31	32	*	*	11
	65-84	298	229	35	13	*	*	*
	85+	34	29	*	*	*	*	*

⁵This data used to be available at <https://hcupnet.ahrq.gov/>, but the interface was changed recently so that data tables with intersections of characteristics are not generated anymore.

⁶J. Vaidya, B. Shafiq, X. Jiang, et al., "Identifying inference attacks against healthcare data repositories," *AMIA Joint Summits on Translational Science proceedings*. AMIA Joint Summits on Translational Science, vol. 2013, pp. 262-266, 2013.

COMPUTING ON ANONYMIZED DATA⁷

- This is an integer programming problem: the values in the table provide constraints for the suppressed values
- We can express each cell as a variable and solve to retrieve concrete values (or at least bounds)

		total # discharges	Race/ ethnicity						
			White	Black	Hispanic	Asian/ Pacific Islander	Native American	Other	Missing
Total# discharges		735	535	82	58	18	1	19	22
Age group	1-17	3	1	x11 [0-2]	x12 [0-2]	x13 [0-1]	x14 [0]	x15 [0-1]	x16 [0-1]
	18-44	70	40	13	x21 [9-10]	x22 [0-6]	x23 [0]	x24 [0-6]	x25 [1-8]
	45-64	330	236	31	32	x31 [10]	x32 [0]	11	x33 [10]
	65-84	298	229	35	13	x41 [2-8]	x42 [1]	x43 [2-8]	x44 [4-10]
	85+	34	29	x51 [1-3]	x52 [1-4]	x53 [0-1]	x54 [0]	x55 [0-1]	x56 [0-1]

⁷J. Vaidya, B. Shafiq, X. Jiang, et al., "Identifying inference attacks against healthcare data repositories," *AMIA Joint Summits on Translational Science proceedings*. AMIA Joint Summits on Translational Science, vol. 2013, pp. 262-266, 2013.

SIMPLE SUPPRESSION OF VALUES IS NOT ENOUGH!

- Adversary may apply algorithms to the output of a privacy mechanism
- Adversary has knowledge about algorithms and resources to apply them at scale

Requirement #4: Post-processing

Post-processing the output of a privacy mechanism must not change the privacy guarantee